NexusOS

AI Business Operating System

Part 03.0

Linear Issue WAL-9

Architecture Design

Project	NexusOS — AI Business Operating System
Part	Part 03.0
Linear issue	WAL-9
Report title	Architecture Design
Generated	2026-07-19
Purpose	Standalone Linear attachment (manual upload)
Source report(s)	docs/reports/Part-03.0/Part-03.0-Architecture-Design.md

This PDF is a presentation export of repository engineering report content. Source report files in the repository were not modified.

Evidence Statement

This report is a retrospective engineering reconstruction produced after implementation. Its contents are derived exclusively from repository evidence. It is not a contemporaneous engineering document. No engineering events are fabricated. Where repository evidence is unavailable, that limitation is explicitly documented.

Table of Contents

Part 03.0 — Architecture Design

- Evidence Statement
- Objective
- Architecture reconstructed from repository evidence
  - Tenancy architecture
  - Domain architecture
  - Index architecture
  - Delete architecture
  - RBAC architecture
- Repository evidence
- Outcome
- Evidence limitations (shared)

Report Content

Content below is copied from the source repository report(s) without alteration.

Part 03.0 — Architecture Design

Linear: WAL-9 **Reconstruction date:** 2026-07-19

Evidence Statement

This report is a retrospective engineering reconstruction produced after implementation.

Its contents are derived exclusively from repository evidence.

It is not a contemporaneous engineering document.

No engineering events are fabricated.

Where repository evidence is unavailable, that limitation is explicitly documented.

Objective

Design the enterprise database architecture: entity relationships, tenant-aware structure, indexing strategy, delete semantics, and RBAC shape.

Architecture reconstructed from repository evidence

Tenancy architecture

- `TenantMixin.company_id` FK → `companies.id` ON DELETE CASCADE
- Child tables carry `company_id` even when reachable through a parent
- `BaseRepository._base_query()` enforces `company_id` filter and raises `TenantIsolationError` if missing

Domain architecture

Entity map documented in `docs/product/database.md` (mermaid ER diagram) covering:

- Identity/RBAC: companies, users, permissions, roles, role_permissions, user_roles
- CRM: customers, contacts, leads, deals, meetings, customer_notes
- Projects: projects, tasks, task_comments, task_attachments
- HR: employees, leave_requests, attendance
- Finance: invoices, invoice_items, payments, expenses
- Inventory: products, warehouses, inventory/stock, suppliers, purchase_orders
- Platform: documents, notifications, audit_logs, activities, AI conversations/messages/reports, workflows/steps

Index architecture

Composite indexes leading with `company_id` for tenant-scoped list hot paths (documented in `database.md`; materialized as 150 indexes in initial migration).

Delete architecture

ON DELETE matrix in `database.md` (company cascade; financial RESTRICT; user-owned SET NULL).

RBAC architecture

```
permissions (global) ← role_permissions → roles (per-company)
                                            ↑ user_roles
                                          users
```

53 seeded `<module>.<action>` permissions; five system roles (Owner/Admin/Manager/Employee/Viewer).

Repository evidence

- `docs/product/database.md`
- `backend/app/models/base.py`, `user.py`, domain model modules
- `backend/app/repositories/base.py`
- `backend/app/db/seed.py`
- Migration `0a961c2ea2cf`
- Commit `838db64`

Outcome

A coherent multi-tenant architecture ready for physical modeling and implementation.

Evidence limitations (shared)

- Part 03.0 code primarily landed in git commit `838db64` (2026-07-09) under the commit message “chore: validate and stabilize backend foundation”, bundled with broader foundation scaffolding. There is no separate contemporaneous commit uniquely titled “Part 03.0”.
- Linear issues WAL-7…WAL-17 were created on 2026-07-18 as a historical execution hierarchy after the schema already existed in the repository.
- No contemporaneous Part-03.0 independent-review finding IDs, meeting notes, or signed review PDF exist in the repository (unlike Part 03.1’s report pack).
- Structural integrity remediations (composite cross-tenant FKs, soft-delete-safe unique indexes, missing `created_by`/`updated_by` FKs) are **Part 03.1** evidence (`f521d38`, revisions `3a2aec52f6d2` / `217e90d00cc0`) and are not attributed as silent Part 03.0 remediations.